Bootstrapping Vision-Language-Action Models for New Robot Embodiments with Staged Reinforcement Learning

WORKING MANUSCRIPT — 26 October 2026
Hardware prose assumes successful completion; measurements remain unfilled.

***Abstract*—Adapting a pretrained vision-language-action (VLA) model to a new embodiment can require learning an action module for the target controller. Sparse task rewards provide little feedback when this module initially produces few successful trajectories. We propose a staged reinforcement learning approach that first uses dense rewards to acquire individual manipulation skills, then refines them with binary success rewards and consolidates successful trajectories through self-imitation. Milestone-based reinforcement learning subsequently composes the skills into a policy conditioned on the complete task instruction. Our approach exposes ordered task progress through phase-aware dense rewards or discrete milestones, with stage-specific credit during composition. We evaluate reward construction with OpenVLA-OFT on LIBERO and the adaptation schedule with a residual action module over pretrained SmolVLA on a cable-driven parallel manipulator. Phase-aware dense feedback improves the trained LIBERO policy under the tested settings; the adapted cable-robot controller achieves 61.43% strict full-task success in simulation development evaluation. *[Success-assumption draft: Physical experiments demonstrate complete object manipulation on the target robot.]* These experiments examine structured feedback for acquiring and composing manipulation skills across different action interfaces.**

***Index Terms*—Robot learning, reinforcement learning, manipulation, vision-language-action models, robot adaptation.**

I. Introduction

Vision-language-action (VLA) models connect visual observations and language instructions to robot commands. Their pretrained representations offer a starting point for manipulation, but their action interface depends on the embodiment used during training. When a new robot requires different action dimensions or control conventions, adapting the model can involve replacing or reinitializing its action head or expert. In this setting, useful visual and language representations may be available before the policy can reliably execute even a basic movement.

Learning a complete instruction directly from binary success is difficult when the initial policy rarely succeeds. Consider "put the apple on the plate": the robot must first approach the apple, grasp it, lift it, carry it, and release it correctly. A failed rollout gives the same final outcome whether the robot never reaches the object or drops it just before placement. Training each component separately can make learning easier, but those components must subsequently work together from a normal task start under one instruction.

Our approach follows this structure. We first use dense rewards to learn individual skills, giving the policy feedback before it can complete them. We then transition to binary skill-success rewards and self-imitation of successful experience. Finally, we train the complete task with staged rewards so that earlier skills and later task progress contribute to learning. The aim is to adapt VLA-based control to a new embodiment through a practical training sequence, rather than to make a particular robot architecture the subject of the method.

We investigate embodiment adaptation using SmolVLA and a cable-driven parallel robot (CDPR), whose gripper is positioned by coordinated cable actuation. This platform serves as a demanding example of a robot with a different mechanical and control structure from conventional serial arms. Its role is to test adaptation under a substantial embodiment change; the approach does not rely on a cable-specific task objective. The evaluated implementation learns a new residual action module while retaining the pretrained SmolVLA action expert and representations (Sec. IV).

We evaluate two aspects of this approach. Our OpenVLA-OFT experiments on LIBERO examine reward construction: phase transitions and newly achieved within-phase progress provide intermediate feedback while preserving the terminal verifier reward. Our CDPR experiments examine embodiment adaptation and full-task composition using discrete milestones and stage-specific advantages. Both expose ordered manipulation progress to learning and evaluate task completion. The reward experiments ask whether intermediate feedback remains useful with a trained action policy; the adaptation experiments ask whether the learning sequence yields complete-task control on the target embodiment.

The paper contributes a dense-to-binary skill-learning schedule with self-imitation and staged composition; phase-aware intermediate feedback with dense-progress and milestone implementations; and evaluations of reward construction and complete-task execution. The complete schedule is instantiated on the CDPR, while LIBERO evaluates the reward design with an established action interface. This distinction defines what each experiment can establish about the approach.

II. Related Work

VLA adaptation. OpenVLA [1] and SmolVLA [2] provide pretrained visuomotor policies with different architectures and deployment costs. OpenVLA-OFT studies efficient adaptation through action chunking, parallel decoding, and continuous-action supervised objectives [3]. Our LIBERO implementation starts from an OpenVLA-OFT checkpoint but applies PPO/GRPO updates to discrete action-token likelihoods. Our embodiment-adaptation implementation freezes the VLA during the final RL phase and learns a bounded residual controller. These parameterizations require different probability models.

Residual and group-relative learning. Probe, Learn, Distill (PLD) combines residual reinforcement learning with autonomous data generation and distillation [4]. Accordingly, residual learning and success-filtered consolidation are established ingredients, rather than standalone novelty claims here. Our emphasis is the training sequence that links skill acquisition, successful-experience consolidation, and complete-task learning. GRPO originates in language-model reinforcement learning [5]; SimpleVLA-RL brings group-relative online optimization to VLA rollouts [6]. Existing binary-GRPO analyses motivate examining which groups actually provide reward contrast [7], [8].

Intermediate rewards and bounded actions. Potential-based shaping provides conditions for preserving optimal policies [9]. The phase-aware reward below is an engineering shaping rule and is not asserted to satisfy those conditions. Likewise, protecting achieved stages is a credit modification rather than a general policy-improvement guarantee. Work on clipped-action gradients establishes the importance of distinguishing a latent sample from its executed clipped value [10]. We apply that distinction to persistent exploration in the CDPR controller and delimit what it implies for the complete optimization procedure.

III. Problem Formulation

We consider a pretrained VLA model adapted to a target robot with action space $\mathcal{A}_e$, where e identifies the embodiment. A policy maps camera observations, proprioception, and a language instruction to robot commands. Its action module must match the target controller; transferring a visual-language representation does not by itself provide a suitable action mapping.

Let $o_t = (I_t^{1:V}, q_t)$ contain the available camera images and proprioception, ℓ denote the language instruction, and h_t collect the observations and controller history available at decision time. The policy generates an action chunk $A_t = (a_t, \dots, a_{t+H-1})$ conditioned on (h_t, ℓ); the controller executes a prefix before replanning. The scene configuration c specifies the episode initialization, including object and target poses. It indexes resets and evaluation groups, rather than an additional textual input to the policy. With binary full-task outcome $Y(\tau)$, the evaluation objective is

$$J(\theta) = \mathbb{E}_{(c,\ell)\sim\rho_{\text{full}}} \mathbb{E}_{\tau\sim P_\theta(\cdot|c,\ell)}[Y(\tau)]. \tag{1}$$

We decompose the task into K ordered skills with binary success predicates Y_k and training reset distributions ρ_k. A skill such as approaching an object has its own instruction and success criterion; it is not equivalent to completing pick-and-place. In particular, placement training from a held-object start does not measure complete manipulation from an empty gripper. The composed task uses a single final instruction and its own reset distribution ρ_{full}.

Dense feedback is used to bootstrap individual skills. Binary predicates then define skill completion, successful experience for imitation, and milestones during full-task RL. A training stage denotes a step in this learning schedule; a task phase denotes progress within one episode. Phase-aware feedback can therefore be dense or milestone-based without changing the schedule itself. Evaluation always measures the declared success event, independently of the training reward. We also distinguish the exploration policy from the deployment policy.

For the CDPR evaluation, the action $a_t \in [-1, 1]^5$ controls Cartesian translation, yaw, and the gripper. Strict full-task success requires grasp, held lift, intentional release, and native placement, without latched carry-slip or wrong-placement events. LIBERO uses its environment goal verifier [11]. These task definitions and scene distributions are different, so their success rates are reported separately.

IV. Learning Skills and Composing the Task

A. Adapting the Action Interface

The target setting is a new embodiment for which the pretrained action interface cannot be used directly. The required intervention can range from a new output projection or controller head to replacement of an action expert. We retain useful VLA representations and learn the action-producing components needed by the new controller. This distinction matters: creating a residual action module does not imply that the complete pretrained expert was reinitialized.

In the evaluated SmolVLA implementation, the pretrained policy, including its action expert and projections, is loaded and retained. Its predicted actions are mapped to the target interface by coordinate selection and normalization. A separate residual MLP is initialized from scratch for initial skill learning; subsequent phases inherit its learned weights. Action-expert LoRA adds trainable low-rank updates in earlier phases while preserving the pretrained base weights. The final continuation freezes the inherited VLA/LoRA and trains the residual and exploration parameters. Thus, this experiment evaluates a newly learned action module over a retained VLA, not a reset of the complete pretrained expert.

B. Dense Rewards for Initial Skill Learning

We divide one object-manipulation task into skills such as approaching the object, picking it up, and transporting and releasing it at the destination. Each skill is initially trained with a dense reward reflecting progress toward its own goal. For example, approach feedback can depend on reduction of gripper-to-object distance, while pickup feedback can distinguish grasp formation and lifting. This gives early rollouts informative feedback even before binary success occurs.

Skill training uses its declared reset distribution and instruction. The task "move to the apple" therefore supplies approach experience; it is not counted as a successful execution of "put the apple on the plate." Similarly, a held-object placement episode teaches transport and release without claiming that the policy performed the preceding grasp. **[To complete: Insert the dense reward functions, skill resets, and training budgets for the selected lineage.]**

C. Binary Refinement and Self-Imitation

After initial skill acquisition, training transitions to a binary reward indicating whether the instructed skill was completed.

This shifts the learning criterion from partial progress to successful execution. We collect successful policy-generated trajectories and use them as supervised targets, retaining experience from earlier skills when consolidating a new one. Here self-imitation means fitting successful experience generated by the policy or a retained policy, not an additional human-demonstration requirement.

For successful skill banks $\mathcal{D}_k^+$, a shared consolidation objective has the form

$$\mathcal{L}_{\mathrm{SI}}(\theta) = \sum_{k=1}^{K} \alpha_k \mathbb{E}_{(o,\ell,a)\sim\mathcal{D}_k^+} \left[\ell_{\mathrm{act}}(\pi_\theta(o,\ell), a) \right], \qquad (2)$$

where α_k controls the skill mixture and ℓ_{act} is the supervised action-fitting loss. In the residual implementation, fitting uses action regression; it need not update every VLA parameter. The policy is subsequently evaluated on each retained skill, since low fitting loss alone does not establish retention.

Historical data also include scripted/oracle-assisted experience, which must be identified separately from policy self-imitation. **[To complete: Specify dense-to-binary switch points, binary RL/SI ordering, bank sizes, mixing weights, and selected checkpoints for each skill.]** These details must follow the actual training record rather than a retrospective assumption that every stage used an identical schedule.

D. Staged RL for Complete-Task Execution

The consolidated skills initialize learning of the complete instruction from ordinary empty-gripper starts. We combine skills by training one policy on the full task with milestone feedback. The final policy must generate the whole sequence under the final instruction; success is not obtained by externally switching between separately prompted specialists. Teacher chains may supply initialization experience, but full-task evaluation does not use teacher assistance.

Milestones credit progress through approach, pickup, and placement while retaining strict task completion as the endpoint. This exposes composition failures that separate skill tests cannot reveal. The complete-task CDPR stage uses sparse milestones, distinct from the dense reward used for initial skill discovery.

Two choices determine how intermediate feedback enters learning: which progress is rewarded, and which actions receive its credit. We implement phase-aware dense feedback for reward-construction experiments and milestone advantages for CDPR composition. The approach, held-pickup, and placement segments correspond conceptually to reach, grasp/lift, and move/place phases; the predicates and return constructions are specified for each implementation. Both distinguish meaningful partial achievements that terminal failure alone would conflate.

E. Phase-Aware Dense Feedback

Our dense implementation represents supported pick-and-place behavior with five ordered phases: reach, grasp, lift, move, and place. Each environment maintains its own tracker from simulator state. If $p_t \in [0,1]$ denotes active-phase progress and B_t the best progress previously reached in that phase, the rewarded increment is $\Delta p_t^+ = \max(p_t - B_t, 0)$. The baseline resets on phase advancement. Revisiting an already reached state within a phase earns no additional progress credit.

Let c_t count newly completed phases and y_t denote terminal verifier success. The additive reward is

$$r_t = \lambda y_t + \mathrm{clip}(w_p \Delta p_t^+ + w_c c_t, -C, C). \qquad (3)$$

Only shaping is clipped; the original terminal reward is retained without a second terminal term. Setting $w_p = 0$ gives phase-only shaping: intermediate credit occurs at phase boundaries. With $w_p > 0$, new progress can earn credit before the next boundary. Thus phase awareness and reward density are separate choices. Both retain achievement information, while our CDPR implementation additionally controls stage-specific credit. **[To complete: Insert exact phase predicates, progress functions, transition/reset logic, and chunk-to-reward aggregation from source code.]**

Rollout selection is a separate mechanism. A group of n trajectories with terminal outcomes Y_i is retained when $L \le n^{-1}\sum_i Y_i \le U$. Bounds excluding zero reject all-failure groups even if dense returns differ. Dense feedback therefore does not automatically remove this filtering bottleneck. Both the reward law and the acceptance rule must be reported when interpreting update throughput.

F. Action Module in the SmolVLA Implementation

A frozen SmolVLA model produces a mapped prior chunk p. The residual input x combines proprioception and frozen visual features; the residual network also receives p. Its bounded mean is

$$\mu_\theta(x,p) = \tanh(p + s_r \tanh f_\theta(x,p)) \,. \qquad (4)$$

Both nonlinearities are retained: s_r bounds the correction before the outer action squashing. The inspected configuration uses $s_r = 1$, eight-slot chunks with replanning every four executed slots, six proprioceptive dimensions, and a 512-dimensional frozen visual projection. The long continuation trains the residual and exploration parameters while keeping inherited VLA/LoRA weights frozen. **[To complete: Confirm these dimensions and controller frequency from the selected run's resolved configuration.]**

G. Group-Relative Updates with Stage Credit

The full-task collector records three binary milestones: approach, pickup, and strict placement, denoted $M_{i1}, M_{i2}, M_{i3} = Y_i$ for candidate i. Within each scene group, milestone returns are centered and divided by their population standard deviation, with numerical flooring and configured advantage clipping. Let A_{ik} denote this advantage and U_{ik} the corresponding usability mask. Constant-return groups provide no milestone contrast. Invalid candidates are excluded from training statistics, although numerical failures remain failures in evaluation.

For approach and pickup records, a terminal bonus supplements local credit:

$$\widetilde{A}_{ik} = U_{ik}A_{ik} + \beta U_{i3}T_\kappa(A_{i3}, M_{ik}), \tag{5}$$

$$T_\kappa(a,m) = \begin{cases} \kappa a, & m = 1,\ a < 0, \\ a, & \text{otherwise.} \end{cases} \tag{6}$$

Placement uses $U_{i3}A_{i3}$ without an additional duplicate bonus. The inspected settings use $\beta = 1$ and $\kappa = 0$: a failed completion does not contribute a negative terminal bonus to an achieved earlier milestone. This protects one credit component, not every parameter supporting the prefix.

With fixed batch advantages, nonnegative stage/candidate weights w_r, and per-record likelihood ratios ρ_r, the clipped surrogate is

$$S(\theta) = \sum_r w_r \min\{\rho_r \widetilde{A}_r, \mathrm{clip}(\rho_r, 1-\epsilon_-, 1+\epsilon_+)\widetilde{A}_r\}. \tag{7}$$

The weights balance retained stage mass and candidate trajectory lengths. Entropy and mean-action penalties accompany the surrogate, and Adam and gradient clipping determine the actual update. Thus the implemented objective is more specific than the sum of three milestone rewards.

H. Conditional Latent-Action Likelihood

At each sampled slot, exploration produces

$$u_t = \mu_\theta(x_t, p_t) + b_t + \sigma_\theta \odot \varepsilon_t, \qquad a_t = \mathrm{clip}(u_t, -1, 1), \tag{8}$$

where ε_t is standard Gaussian and b_t is a gated contribution from an exogenous episode-level offset. The collector retains the unclipped sample u_t, the realized offset, and the old log probability. New/old ratios score the same sample under the conditional Gaussian density $\mathcal{N}(\mu_\theta + b_t, \mathrm{diag}\,\sigma_\theta^2)$.

Scoring the clipped action as a Gaussian observation incorrectly represents probability mass at the bounds. Treating a shared offset as fresh independent noise at every step also loses temporal dependence. Conditioning on its recorded realization avoids both mismatches. It does not eliminate biases introduced by stage selection, invalid-candidate filtering, or repeated surrogate optimization.

V. Mechanism Analysis

A. Latent Scores and Binary Contrast

Assume fixed dynamics, a frozen prior, an exogenous offset law, and regularity allowing differentiation under the expectation. The augmented trajectory density then factors into exogenous terms and conditional latent-policy terms. Consequently,

$$\nabla J = \mathbb{E}\left[Y \sum_t \nabla \log \pi_\theta(u_t \mid h_t, p_t, b_t) \right]. \tag{9}$$

To see this, differentiate the trajectory density: only the policy factors depend on θ at fixed recorded history, while clipping is a deterministic execution map. The offset density cancels in likelihood ratios. A learned covariance contributes its own Gaussian score. Equation (9) describes the full-trajectory score objective, not an unbiasedness guarantee for the stage-filtered PPO surrogate [10].

For G conditionally independent candidates of one scene with binary milestone probability p, the probability of a nonconstant group is

$$P_{\mathrm{mixed}} = 1 - p^G - (1-p)^G. \tag{10}$$

This follows by excluding the all-success and all-failure outcomes. At $G = 8$, changing p from 0.03 to 0.30 increases this probability from 21.6% to 94.2%. Thus milestones can supply contrast when completion is rare, although an almost certain milestone also loses contrast. For heterogeneous scenes, average the scene-specific probabilities rather than inserting the overall success rate. These observations agree with binary-group analyses [7], [8]. Our feedback implementations expose contrast at different resolutions: milestones distinguish achievement levels, while dense progress can distinguish trajectories within one level. Neither guarantees useful updates: progress must vary, and rollout filtering must retain the group. In particular, a terminal-accuracy filter can discard an all-failure group even when its dense returns differ.

B. What Prefix Protection Guarantees

Consider a benchmark of three sequential Bernoulli gates with separate logits, $q_k = \mathrm{sigmoid}(\theta_k)$ and $J = q_1 q_2 q_3$. A reached gate produces one decision X_k, with score $X_k - q_k$. Use sign-preserving binary group normalization, nonnegative weights, on-policy raw ascent, $\beta \geq 0$, and $\kappa = 0$, without regularization or optimizer momentum. Then every realized batch direction has $d_k \geq 0$, and

$$J(\theta + \eta d) \geq J(\theta), \qquad \eta \geq 0. \tag{11}$$

Indeed, an achieved gate receives nonnegative milestone credit and no negative completion bonus; a failed gate receives nonpositive credit. In either case credit and score have the same sign. Their weighted product is nonnegative, so every gate logit can only increase. This is a sufficient explanation of the protection rule, not evidence that symmetric credit must fail in this model.

The neural controller shares parameters across stages and assigns many actions to each stage, so the sign argument no longer applies. For an objective with an L-Lipschitz gradient on the actual update segment,

$$J(\theta + \Delta) - J(\theta) \geq \nabla J^\top \Delta - \tfrac{L}{2}\|\Delta\|^2. \tag{12}$$

The inequality follows by integrating the gradient along that segment. Positive alignment and a sufficiently small update would imply improvement, but neither an alignment bound nor L has been measured for the CDPR policy. PPO clipping alone does not establish this condition [12]. The analysis also gives no automatic guarantee for hardware transfer or for deployment with a different exploration law.

VI. Experimental Setup

Our evaluation examines reward construction on LIBERO and embodiment adaptation on the CDPR. The former compares phase-aware dense feedback with terminal-only and

phase-only configurations; the latter evaluates complete-task control after the proposed learning schedule. Model interfaces, scene distributions, and success verifiers are specified for each setting.

A. New-Embodiment Adaptation with SmolVLA

The adaptation study uses a cable-driven manipulator as an unconventional embodiment. The simulation implements a five-channel CDPR in MuJoCo Warp/MJ-Lab, camera inputs, four object categories (apple, orange, potato, tomato), and plate/bowl destinations. The matched evaluation uses 512 distinct development scenes, excluding the 128-scene in-run validation panel, with four repeats per checkpoint. Each episode has a 128-decision horizon and no settling decisions or teacher/yaw assistance. Deployment uses the residual mean with VLA prior noise scale 1. Numerical failures remain in all-attempt denominators. **[To complete: Supply simulator version, scene-generation ranges, camera calibration, and the resolved action timing.]**

The main comparison evaluates the 20M and 40M latent-likelihood bounded-residual checkpoints, with a panel-selected 39.8M checkpoint as a secondary comparison. These counters measure cumulative selected actions since latent-likelihood conversion; they exclude earlier inherited training and do not count all sampled interactions. The 20M–40M leg adds 19,987,881 selected actions, 609,792 episodes, and 299 updates over 89.14 recorded training hours. The learning rate remains 10^{-5} and the recorded LoRA change is zero.

Reported effect intervals use a percentile bootstrap of per-scene mean differences, keeping repeats clustered by scene. Within-scene permutation p-values use the recorded Holm correction for the primary strict comparisons. Subgroup results are exploratory. This is one continuing training lineage; four evaluation repeats are not four independent training seeds. Final-test results and controlled comparisons of dense versus binary skill starts, self-imitation on/off, and staged versus terminal-only full-task learning remain to be added. Likelihood and prior-contribution ablations assess implementation choices separately.

B. Reward Construction on LIBERO

We test whether task-progress feedback benefits a trained policy when complete-task outcomes provide limited contrast, using the phase-aware reward in Sec. IV-E. These experiments evaluate reward construction with an established action interface.

We train OpenVLA-OFT in SimpleVLA-RL on the ten LIBERO-Spatial tasks, using two NVIDIA L40 GPUs and an aggregate rollout batch of 32. LoRA rank and alpha are both 16, applied to the language-model projector; gradient checkpointing and FSDP offloading reduce memory demand. These updates optimize discrete action-token likelihoods, despite the continuous-action supervised formulation associated with OFT. **[To complete: Confirm implementation commit, tokenizer/action decoding, adapted modules, group size, and optimization schedule.]**

All aggregate reward configurations start from 61.0% validation success. The primary dense condition uses $(w_p, w_c, C, \lambda) = (0.2, 0.3, 0.05, 5)$; its terminal-only control disables shaping and retains $\lambda = 5$. The filtered dense condition additionally uses $[L, U] = [0.1, 0.9]$. With $\lambda = 1$, dense shaping uses $w_p = w_c = 0.05$, while phase-only shaping uses $w_p = 0, w_c = 0.2$; both use $C = 0.05$.

We report the final recorded evaluations, balanced across the ten tasks. To examine low-success learning, we restrict training and validation to `task_1`, starting at 10%, with $[L, U] = [0.1, 1.0]$ in both conditions. **[To complete: Complete evaluation counts, seeds, canonical task name, elapsed budgets, and accepted-update counts.]** Percentages are descriptive until these records support uncertainty estimates and budget matching.

C. Physical Deployment

[Success-assumption draft: The following protocol and result prose are prepared for a completed successful experiment; retain only details confirmed by the actual trial record.] We evaluated the selected policy on the physical cable-driven platform using **[To complete: hardware configuration]**, **[To complete: camera/state-estimation setup]**, and a control frequency of **[To complete: Hz]**. The deployment condition was **[To complete: direct transfer, calibration only, or hardware adaptation; describe exactly]**. The hardware checkpoint was **[To complete: path and hash]**; it must not be inferred from the simulation table.

Each trial began with an empty gripper and an object outside the destination. A successful trial required acquisition, sustained lift and transport, deliberate release, and placement in the instructed receptacle without assistance. We recorded all attempts, interventions, and aborts under **[To complete: trial/reset protocol and stopping rule]**. The evaluation comprised **[To complete: objects, destinations, layouts, sessions, and trials per policy]**, with **[To complete: baseline and ordering]**. The earlier outline's 80-trial design is only a suggestion, not a completed sample size.

VII. Results

A. Complete-Task Performance After Embodiment Adaptation

The selected SmolVLA-based controller performs the full manipulation instruction on the target embodiment. Its 40M endpoint completes 1,258 of 2,048 attempts (61.43%), compared with 1,056 (51.56%) at 20M under the same development protocol (Table I). The gain is 202 successes, or 9.86 percentage points, with a scene-paired 95% interval of [7.37, 12.30] and Holm-adjusted $p = 0.00010$. Counts are compared before rounding. This documents improvement with continued training, rather than an isolated causal effect of the likelihood change.

The panel-selected 39.8M checkpoint achieves 1,256/2,048 (61.33%). Its direct difference from final40m is -0.10 points, with a paired interval of $[-2.05, 1.81]$ and $p = 0.9598$. Thus its higher in-run peak does not provide a demonstrated advantage on the separate development scenes; this is not

TABLE I
MATCHED CDPR DEVELOPMENT EVALUATION: 512 SCENES × FOUR REPEATS PER CHECKPOINT. NF DENOTES NON-FINITE EPISODES. ALL RATES INCLUDE FAILURES IN THE DENOMINATOR.

Checkpoint	Strict count	Strict (%)	Native (%)	NF (%)
20M	1056/2048	51.56	56.74	1.32
40M final	1258/2048	61.43	66.46	1.37
39.8M peak	1256/2048	61.33	66.89	1.76

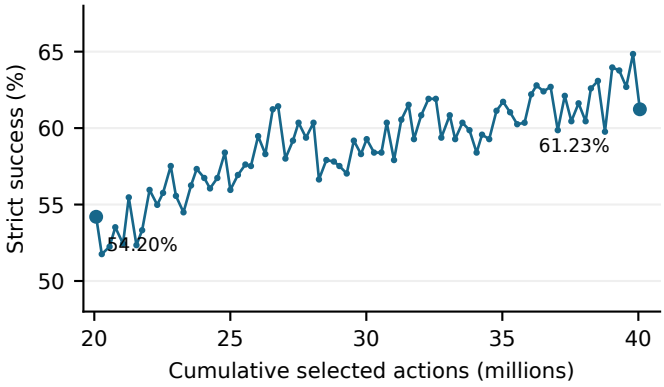


Fig. 1. The 20M–40M continuation on the repeated in-run validation panel (1,024 episodes per reading). These correlated readings describe one training lineage; they are distinct from Table I's 512-scene matched comparison. The horizontal axis counts selected actions since conversion, excluding inherited training.

an equivalence test. Figure 1 shows the unsmoothed in-run trajectory separately from the matched endpoint comparison.

The retained earlier 2M pilot provides additional development context: the bounded-residual continuation achieves 843/2,048 (41.16%) versus 756/2,048 (36.91%) for its source, a paired gain of 4.25 points [2.10, 6.40]. These earlier repeats are not combined with the 20M/40M comparison to infer a single paired learning curve.

B. Stage Retention and Remaining Failures

Continued CDPR training improves physical-grasp and held-lift rates from 85.84% and 80.52% to 89.65% and 85.89%. Strict completion conditional on held lift rises from 64.04% to 71.52% as a descriptive ratio. Carry-slip events decrease from 28.32% to 23.83%, and wrong-placement events from 25.00% to 19.63%. Their paired difference intervals are $[-6.79, -2.25]$ and $[-7.62, -3.17]$ points, respectively. Event flags may overlap and must not be stacked into an exclusive failure partition.

Plate and bowl strict rates at 40M are 68.58% and 54.22%. The far-side target-y quartile improves from 10.08% to 29.07%, but remains substantially weaker than the nearer quartiles (Fig. 2). Potato remains the weakest object category at 46.11%. These exploratory breakdowns locate residual difficulty; they do not establish robustness to new objects or workspace distributions.

Non-finite episodes occur in 1.32% and 1.37% of the 20M and 40M evaluations. The difference interval $[-0.73, 0.83]$ points does not establish equivalence or absence of numerical

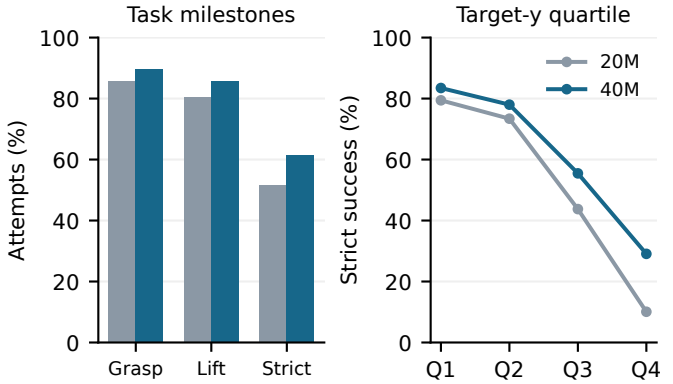


Fig. 2. Matched development diagnostics for 20M and 40M: (left) marginal grasp, held-lift, and strict outcomes over all 2,048 attempts; (right) strict outcome by target-y quartile, with 130, 125, 128, and 129 scenes, each repeated four times. These are descriptive rates; confidence intervals for the main paired effect appear in the text.

failures. In the earlier 2M pilot, adding a separate correction network yields 37.55% strict success and 3.56% non-finite episodes, with no demonstrated strict gain over its source: 0.63 points $[-1.76, 3.03]$. This arm is not the architecture producing the 61.43% result.

The central schedule requires matched comparisons of dense versus binary rewards at skill initialization, binary refinement after dense learning, self-imitation with and without retained skill data, and staged versus terminal-only full-task training. These comparisons remain outstanding. Protected-credit, likelihood, and action-prior controls address secondary implementation choices; continuation results cannot substitute for either set of ablations.

Skill consolidation must also be evaluated separately from full-task completion. Retained campaign reports document recovery of earlier instruction families with balanced self-imitation, but also unsuccessful SFT attempts on an already trained full-task policy. We therefore treat self-imitation as a consolidation step whose benefit depends on the experience bank, rather than assuming that every additional imitation pass improves the final task. **[To complete: Add matched per-skill before/after consolidation results and exact bank provenance.]**

C. Phase-Aware Feedback on LIBERO

Our LIBERO reward comparison appears in Table II. Primary phase-aware dense feedback increases validation success from 61.0% to 65.3%, compared with 61.7% for terminal-only reward: a 3.6-percentage-point endpoint difference. Filtered dense reaches 64.7%. Phase-only shaping reaches 62.3%, whereas weaker-terminal dense remains at 61.0%. The primary dense and phase-only rows differ in terminal coefficient and shaping weights; their 3.0-point gap does not isolate within-phase progress. Likewise, the table does not isolate terminal scaling. The observed ordering supports structured intermediate feedback under the tested settings, rather than a general claim that denser reward is better.

In our low-success experiment, terminal-only and dense training reach 11% and 18%, respectively, from the same

TABLE II
LIBERO-SPATIAL REWARD COMPARISON. ALL CONFIGURATIONS START AT 61.0%; Δ IS IN PERCENTAGE POINTS. EVALUATION COUNTS, SEEDS, AND UNCERTAINTY REMAIN TO BE COMPLETED.

Configuration	Success (%)	Δ
Primary dense, $\lambda = 5$	65.3	+4.3
Filtered dense $[0.1, 0.9]$	64.7	+3.7
Phase-only, $\lambda = 1$	62.3	+1.3
Terminal-only, $\lambda = 5$	61.7	+0.7
Dense, $\lambda = 1$	61.0	0.0

TABLE III
LOW-SUCCESS LIBERO EXPERIMENT. BOTH CONDITIONS USE GROUP-ACCURACY FILTERING $[0.1, 1.0]$. VALUES ARE PERCENTAGES; EVALUATION TIMESTAMPS AND DENOMINATORS REMAIN TO BE COMPLETED.

Evaluation	Terminal-only	Dense
Initial	10	10
Intermediate	8	12
Final	11	18

10% initialization (Table III). Dense training also assembles accepted batches faster and completes more optimizer updates during the observed interval. **[To complete: Quantify this training-log observation using timestamps, attempt counts, and matched budget endpoints.]** It does not yet establish a numerical speedup or improvement per optimizer update.

These results support our use of intermediate feedback after basic skills exist, as in CDPR task composition. They do not determine whether dense shaping or milestone credit is preferable on the CDPR, nor establish the benefit of binary refinement or self-imitation. Those comparisons test other parts of the adaptation schedule.

D. Finite-Model Mechanism Checks

Independent finite-model enumeration verifies the binary-group identities for $G = 2, 4, 8$ at seven probabilities with maximum absolute error 1.2×10^{-15}. All 336 enumerated sequential-gate groups at $G = 2, 3, 4$ satisfy the protected-credit sign property; checks over three parameter vectors and three raw-ascent step sizes preserve success. These tests validate the stated finite models, not the neural robot policy.

In a two-decision toy process with persistent offsets, the conditional latent score agrees with the analytic and finite-difference derivatives. At $\theta = 0.1$, the derivative is 0.3170, whereas treating exact one-step marginals as independent yields 0.6344. Policy-record likelihood audits, measured mixed-group frequencies, and independent update-alignment measurements are still needed to connect these mechanisms to the reported robot improvement.

E. Real-World Manipulation

[Success-assumption draft: This subsection is written under the author's assumption of successful completion. It is not a report of verified hardware outcomes.] The deployed controller completed the instructed pick-and-place sequence on the physical cable-driven robot, progressing from an empty-gripper start through acquisition, lift, transport, and deliberate release. Figure 3 reserves the apparatus and time-ordered rollout panels. The demonstrated behavior establishes feasibility on the tested apparatus; its reliability is quantified by the complete trial inventory in Table IV.

TABLE IV
PHYSICAL EVALUATION SLOTS. ALL MEASUREMENTS ARE PENDING; DASHES ARE NOT ZERO OUTCOMES. TRANSFER/ADAPTATION CONDITION AND SAMPLE SIZES MUST ACCOMPANY THE COMPLETED TABLE.

Policy	Successes/trials	Interventions	Time (s)
Selected controller	—	—	—
Matched baseline	—	—	—

(a) Physical CDPR setup
Insert `figures/hardware/setup.jpg`

(b) Recorded rollout: approach, grasp, lift, transport, release
Insert `figures/hardware/sequence.jpg`

Fig. 3. Reserved physical experiment panels. Add actual photographs, frame timestamps, instruction, object/destination, trial ID, and deployment condition. The intended successful sequence is part of the success-assumption draft and remains unverified.

Across **[To complete: N]** attempts, the selected policy achieved **[To complete: S]** strict successes (**[To complete: rate and 95% interval]**). The baseline achieved **[To complete: successes/trials under the same conditions]**. We observed **[To complete: interventions and aborts]**, and successful trials required **[To complete: completion-time summary]**. The dominant residual failure was **[To complete: measured category and count]**; **[To complete: failure sequence/video]** documents its manifestation. These quantities must come from the trial log, including unsuccessful attempts.

The simulation and hardware outcomes use corresponding task-completion criteria, but differ in sensing, contacts, actuation, and reset variability. The transfer description must specify calibration and any hardware adaptation. A successful sequence alone does not establish zero-shot transfer, a particular success percentage, or superiority to the baseline.

VIII. DISCUSSION AND LIMITATIONS

The central question is how to adapt VLA-based control when an existing action interface does not provide useful behavior on a new robot. Dense reward supports skill discovery, binary refinement targets completion, self-imitation consolidates experience, and staged RL learns the complete instruction. These roles require separate measurements of acquisition, retention, and composition.

Our experiments examine a common mechanism: ordered task achievements distinguish trajectories that share the same terminal failure. CDPR exposes these distinctions through milestone advantages; LIBERO also resolves progress within a phase. The observed dense advantage supports intermediate feedback beyond initial acquisition. Its granularity remains an empirical choice: our LIBERO ablations do not isolate progress density, and a matched dense-versus-milestone CDPR comparison remains outstanding.

Binary skill refinement and milestone composition serve different objectives and reset distributions. Our schedule uses binary rewards to target skill completion, then reintroduces structured feedback for the longer composed task. Our LIBERO results support this rationale; they do not establish that switching to binary rewards, or the full schedule, is superior to continued dense training.

The CDPR is one embodiment-adaptation case, using a new residual over a retained SmolVLA expert. Resetting an entire expert or applying the full schedule to additional embodiments remains untested; LIBERO starts with a trained policy. Skill consolidation also gives no guarantee that further SFT on full-task successes helps: unsuccessful campaign attempts show the importance of data mixture, coverage, and action consistency.

The current CDPR endpoint uses development scenes and one training lineage. Matched schedule, self-imitation, and composition ablations, independent seeds, and a locked final test remain necessary for stronger attribution. Our LIBERO evaluation requires completed trial denominators and quantitative throughput records. Physical success remains a drafting assumption until trial evidence is inserted. Manual rewards, historical scripted initialization, calibration, and inherited training costs remain part of the method's scope.

IX. Conclusion

We proposed a training sequence for adapting VLA-based control to new robot embodiments: dense-reward skill acquisition, binary-reward refinement with self-imitation, and staged RL for complete-task execution. Our phase-aware feedback exposes intermediate progress through dense rewards or milestones. LIBERO experiments show an observed benefit of dense feedback with a trained action policy; the adapted SmolVLA controller achieves 61.43% strict full-task success in CDPR simulation development evaluation. *[Success-assumption draft: Physical experiments demonstrate execution of the complete manipulation task on the target robot.]* Controlled schedule comparisons and further embodiments are needed to establish the contribution and generality of each training stage.

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